

Chapter 80 — Surgery of Testicular Tumours

Campbell-Walsh-Wein Urology · Part VII Male Genitalia · Chapter Notes

MANAGEMENT OF THE TESTIS MASS

History and Physical Examination, Ultrasonography, and Preorchidectomy Evaluation

- Any testicular mass needs **prompt, complete evaluation**, especially at peak-incidence ages: the cancer is often **rapidly progressive**, and delay raises the intensity and morbidity of curative therapy.
- Four components: **history** (growth rate, symptoms) → **examination** → **scrotal ultrasound** → **serum tumour markers**.
- Examination covers the **supraclavicular nodes, breasts and abdomen**, then the scrotal contents and the mass itself.
- Ultrasound is **not mandatory** but characterises the mass and records laterality; its **key role is the contralateral testis (bilateral synchronous masses about 1%)**.
- **Serum tumour markers (STMs)** support the diagnosis of germ cell tumour (GCT), give staging information and set the **post-orchidectomy baseline**.
- A **testicular prosthesis** should be discussed **before** surgery.

Radical Orchidectomy

- For suspected testicular malignancy, **radical orchidectomy is the diagnostic and therapeutic treatment of choice**.
- **Inguinal approach**, giving **early control of the spermatic cord**, then removal of testis, epididymis and cord to the internal inguinal ring.

Technique

- **Set-up:** supine. Prepare from above the umbilicus to mid-thigh, and the genitalia back to the perineum. Drape to expose the scrotum, ipsilateral anterior superior iliac spine and pubic tubercle. Examination under anaesthesia locates the external ring.
- **Sequence:** **3–5 cm transverse incision along Langer lines** over the canal (extend hockey-stick over the scrotum, or angle caudally from the outset, if needed) → incise **external oblique fascia along the canal for about 4 cm** (Weitlaner or Gelpi retractor) → **identify and preserve the ilioinguinal nerve** on the anterior cord → mobilise the cord at the pubic tubercle and encircle it with a Penrose drain → divide external spermatic fascia and cremaster → deliver the testis with upward pressure on the hemiscrotum → divide the gubernaculum → dissect the cord to the peritoneal reflection at the internal ring.
- **Vas and gonadal vessels are ligated and divided separately with non-absorbable suture, leaving a 1–2 cm tail**, so the gonadal stump can be found and excised at a later retroperitoneal lymph node dissection (RPLND). Some surgeons drop the stump preperitoneally, deep to the internal ring. Because the vas is separated, it need not be removed at RPLND.
- **Closure:** irrigate and check haemostasis → lay the ilioinguinal nerve back in the canal → close the external oblique aponeurosis → subcutaneous tissue in 1–2 layers → skin → **scrotal support with fluffed dressings for 2–3 days** to limit swelling and haematoma.

Partial Orchidectomy

- **Rationale:** with survival now the norm, the aim is **less long-term treatment morbidity** without lost efficacy.

- **Indications (all three, highly selected):** tumour is **polar**, ≤ 2 cm, and the **contralateral testis is compromised or absent**. It is justified only when the **risk of anorchia outweighs the risk of local recurrence**.
- If malignancy is uncertain, perform inguinal exploration with early cord control and **excisional biopsy $\pm$ frozen section**.
- With a **normal contralateral testis**, elective testis-sparing surgery **should not be attempted**.
- **Emerging: very small masses (<5 mm) with negative markers** have a low rate of malignancy. They may be candidates for partial orchidectomy or, highly selectively, **close surveillance**.

Technique

- **Sequence:** same as radical until delivery $\rightarrow$ isolate the testis with towels $\rightarrow$ **intraoperative ultrasound** to localise $\rightarrow$ **hypothermia can be omitted if resection time is <30 minutes** $\rightarrow$ incise the tunica albuginea sharply $\rightarrow$ excise the mass with a **rim of normal seminiferous tubules** (negative margin) $\rightarrow$ close the tunica with absorbable suture $\rightarrow$ **three-point fixation** in the dependent scrotum (gubernaculum and septum).
- **Incision orientation: vertical (long axis) for a ventral-midline approach vs horizontal for lesions medial or lateral to the midline.** This spares the **segmental arteries** running just deep to the tunica.
- **Intratubular germ cell neoplasia** in the surrounding parenchyma is debated: **completion radical orchidectomy vs adjuvant radiotherapy to the remnant**. The surgeon's view of this risk decides whether frozen section is used.
- Traditionally, confirmed GCT received **18–20 Gy adjuvant radiotherapy**. Spermatogenesis is then permanently lost, so in a solitary testis the **main benefit of sparing is Leydig cell (androgen) function**.
- Selected patients can be **observed without radiotherapy**, even with carcinoma in situ: a meta-analysis of case series showed **low local recurrence (about 7.5% at 3–5 years)**. More may be observed in future; this needs study.
- **Not debated: ipsilateral local recurrence $\rightarrow$ completion radical orchidectomy**, regardless of prior adjuvant therapy or residual testicular function.

Delayed Orchidectomy

- A minority present with **diffuse metastatic and/or symptomatic GCT** that needs immediate chemotherapy. Diagnosis is by **biopsy of a metastatic site**, or presumptively from clinical features $\pm$ markers.
- **Delayed radical orchidectomy is recommended for all such patients, whatever the response elsewhere**, because chemotherapy **response is discordant between the testis and other sites**.
- An **apparent retroperitoneal/extragonadal primary** is more controversial. Testicular biopsy shows intratubular germ cell neoplasia in **up to 42%**, and **about 5% develop metachronous testicular cancer** if observed after chemotherapy.
- **Retroperitoneal disease that lateralises** in a testicular-primary pattern $\rightarrow$ **orchidectomy on that side**. Post-chemotherapy specimens mostly show teratoma or necrosis, i.e. a "**burned-out primary**" or complete response.
- If observation is chosen: **monthly testicular self-examination** plus regular clinician examination.

Postorchidectomy Evaluation

- Clinical stage = **pathology + imaging + markers**.
- **Contrast-enhanced computed tomography (CT)**, with intravenous and oral contrast, images the retroperitoneum; **magnetic resonance imaging (MRI)** is an alternative.

- **Fluorodeoxyglucose positron emission tomography (FDG-PET) and lymphangiography have no important role** in initial diagnosis and staging.
- **Repeat STMs after surgery: α -fetoprotein (AFP), β -human chorionic gonadotropin (β -hCG) and lactate dehydrogenase (LDH).** The **trend** guides subsequent management.

KEY POINTS: MANAGEMENT OF THE TESTIS MASS

- Suspected malignancy → **inguinal radical orchidectomy**.
- **Partial:** polar, ≤ 2 cm, compromised/absent contralateral testis.
- Chemotherapy-first presentation → **delayed orchidectomy in all**.

Concept — Testis-mass surgery is built to **avoid violating the scrotum and to leave a findable cord stump for later staging surgery**. Organ sparing is the exception, allowed only when anorchia is the greater harm, and the testis is always removed eventually because it responds to chemotherapy differently from metastases.

RETROPERITONEAL LYMPH NODE DISSECTION

- Spread is to the **retroperitoneal nodes in a predictable, side-specific pattern**; the **technique is broadly the same** in every setting.
- **Retroperitoneal node regions (Fig. 80.1):** 1 paracaval · 2 precaval · 3 interaortocaval · 4 pre-aortic · 5 right iliac · 6 para-aortic · 7 left iliac · 8 inter-iliac · 9 right suprahilar · 10 left suprahilar.

TERMINOLOGY OF RPLND BY TIMING AND PRIOR THERAPY

Term	Setting	Typical indication
Primary RPLND	After orchidectomy, no chemotherapy	High-risk clinical stage I or low-volume clinical stage II (N1) nonseminomatous GCT (NSGCT) with normal STMs
Post-chemotherapy RPLND (PC-RPLND)	After induction chemotherapy	Usually a residual mass >1 cm with normal post-chemotherapy STMs (some debate)
Salvage PC-RPLND	After induction and salvage chemotherapy	Residual disease after second-line therapy
Desperation PC-RPLND	After chemotherapy	STMs still elevated
Reoperative (redo) RPLND	After a previous RPLND	Retroperitoneal recurrence
Resection of late relapse	>24 months after complete response to primary chemotherapy	Late retroperitoneal relapse

- The terms are defined by **what has already failed**: chemotherapy lines (salvage), markers (desperation), surgery (redo) or time (late relapse). Each step down means **more chemoresistant disease and a harder operation**.

Preoperative Planning

- Review the CT for **mass location and anatomical variants** (e.g. duplicated renal vessels), avoiding unplanned intraoperative consultations.
- **Timing:** CT within **6 weeks** and STMs within **7–10 days** of surgery. **No bowel preparation**.
- **Sperm banking** is offered and recommended to all who want future paternity.
- **Bleomycin:** pulmonary toxicity occurs in **about 11% after bleomycin-based first-line chemotherapy vs 1.7% without**, and grade 3–4 toxicity is **fivefold** higher. So: **pulmonary function tests** → any deficit to a

pulmonologist → alert anaesthesia and all perioperative staff → **low fraction of inspired oxygen (FiO₂) and conservative fluids** → confirm the plan directly at the surgical time-out.

- **Chest CT** if there are prior lung lesions or simultaneous lung resection is planned.
- **Inferior vena cava (IVC) thrombus: total occlusion** → IVC resection; **incomplete occlusion** → **allograft reconstruction** as needed.

Surgical Technique

- **Open RPLND:** supine, arms in a T slightly below 90°. An **orogastric tube** is almost always enough. **Midline incision from xiphoid to just below the umbilicus.**
- **Transperitoneal (traditional) approach:** enter sharply → divide the round ligament → release the falciform ligament towards the superior IVC, avoiding hepatic injury from retraction.
- **Extraperitoneal approach** (selected patients): same incision, but the peritoneum is peeled off the fascia starting infraumbilically, so the **bowel stays contained**. It may **shorten stay and speed return of bowel function**.

Exposure of the Retroperitoneum

- **Sequence (Fig. 80.2):** self-retaining retractor → incise from the **tip of the caecum to the medial side of the inferior mesenteric vein (IMV)** → for full mobilisation with bowel on the chest, incise the **white line of Toldt up the ascending colon to the foramen of Winslow** → for large left-sided tumours, **divide the IMV** → for a modified left approach, incise the white line of Toldt of the descending colon.
- Develop the plane between mesentery and retroperitoneal fat, **staying anterior to the gonadal vein and ureter**. Free the duodenum and inferior pancreas off the renal vessels, IVC and aorta. **Liberal clips here reduce postoperative chylous leak.**

Split and Roll Technique (Video 80.1)

- **Top-down** start over the **left renal vein or IVC; ligate the cephalad lymphatics at the renal veins**. Mass size often dictates the order.
- Starting on the aorta: account for the **postganglionic sympathetic nerves on the anterior aorta** and the **superior hypogastric plexus**, which the right-sided fibres lead to.
- **Split at 12 o'clock on the aorta just below the left renal vein** → carry caudally to the **inferior mesenteric artery (IMA)** (Fig. 80.3). In nerve sparing, **identify and preserve the postganglionic nerves at or before the IMA** before going further caudally.

Left Para-Aortic Packet

- Template: roll the descending colon medially (white line of Toldt). Bilateral: through the mesenteric root, small bowel cephalad.
- Divide the **left gonadal vein** where it crosses the ureter, or move it medially. Retract the ureter laterally.
- Split along the aorta and iliac vessels caudally to the **ureteric crossing (about half of the left common iliac artery)** → roll laterally → **doubly ligate the lumbar arteries up to the renal hilum**.
- There are **usually three pairs of lumbar arteries**, evenly spaced from hilum to common iliacs. The **second pair arises at the level of the IMA**, which makes it a landmark.

Interaortocaval Packet

- **Right nerve-sparing:** split-and-roll on the anterior IVC, ligating the gonadal vein. **Non-nerve-sparing, starting on the aorta:** control the medial aorta first.
- Caudally along the IVC to the iliac bifurcation, then up the right common iliac artery to the ureteric crossing, where the right gonadal vein is ligated and the ureter retracted.

- **Lumbar veins (usually 2–3, variable):** a **superior right lumbar vein** just below or behind the renal veins, and a central **common lumbar vein, the largest**, which may cross **beneath the aorta** and then need para-aortic ligation as well.
- **Relationship at the IMA level:** the **IMA, the common lumbar vein and a postganglionic sympathetic nerve** (second infrarenal lumbar splanchnic nerve) crossing onto the anterior aorta. Identify the nerve prospectively and dissect it free.
- **Sequence:** control the medial IVC → control the medial aorta → roll the tissue medially → **ligate the three medial lumbar arteries** (Fig. 80.4) → release the lymphatics cephalad towards the **crus and anterior spinous ligament at the right renal artery and vein, ligating the cephalad lymphatics** (chylous leak) → proceed caudally over the anterior spinous ligament to the left common iliac vein.
- Watch lumbar vessels entering the body wall; identify the **right renal artery and any accessory arteries**.

Right Paracaval Packet

- **Smallest and easiest;** tapers to almost nothing at the right renal vessels. In bilateral split-and-roll, release it at the right renal artery and pass it **beneath the IVC en bloc with the interaortocaval packet**.
- Release it from psoas, **sparing the genitofemoral nerve**, down to the right common iliac vessels and ureteric crossing.
- The **right sympathetic trunk** lies posterolateral to the IVC. It is **often mistaken for lymphatic tissue or a vessel** and should be preserved, especially in nerve sparing.

Gonadal Vein

- Incise the overlying peritoneum, sweep the ureter off posteriorly (a **left vein reached through the mesentery passes under the sigmoid**), then resect with gentle traction **to the internal ring and the orchidectomy suture**.

Nerve Sparing

- Antegrade ejaculation depends on **four postganglionic sympathetic efferents (L1–L4)**, which vary. Surgically they often fuse into **about three right fibres emerging beneath the IVC** and **about three left para-aortic fibres**. Both run over the **anterior aorta to the superior hypogastric plexus** (Fig. 80.5).

LEFT- VS RIGHT-SIDED POSTGANGLIONIC FIBRES

	Left	Right
Where found	Lateral border of the aorta , emerging beneath the left renal vein and around the left common iliac artery	During precaval dissection and medial isolation of the lumbar veins
Relations	Behind the left renal vein and first lumbar artery ; the lower fibres lie by the left gonadal artery, heading to the IMA	Emerge from beneath the IVC , running obliquely to the plexus beside the right superior lumbar vein and common lumbar trunk , so they are at risk when these veins are ligated

- Sweep tissue off the fibres by sharp or blunt dissection and tag them with vessel loops if needed. Once all fibres are isolated, **pass the packet sequentially around the web of fibres** as it is released from the body wall (Fig. 80.4: A right fibres, B left fibres, C right nerve sparing with lumbar arteries ligated).

Closure and Postoperative Care

- Inspect for residual tissue, lymph leak and bleeding; **clip or ligate leaks**; warm irrigation reveals vessels in spasm.
- Optional: reapproximate the **posterior parietal peritoneum** (bowel adhesion to the great vessels) and the **mesenteric root** (volvulus). Always inspect the bowel, liver, stomach and pancreas.

- **No routine drain.** Consider one after large retroperitoneal, retrocrural or bowel resection, where third-spacing is large: a **non-suction drain**, removed when output is **serous and <100 mL/24 h on a regular diet**.
- **No nasogastric tube → clear fluids the evening of surgery → low-fat diet on days 1–2. Non-opioid analgesia** (ketorolac, paracetamol, gabapentin) reduces opioid use and bowel dysfunction, and possibly stay. **Mobilisation mandatory on day 1. Discharge at 3–5 days (target day 3).** Individualised care overrides a standardised enhanced-recovery pathway.

Concept — RPLND is a **landmark-driven vascular operation**. Lymphatics are ligated at the cephalad limit (to prevent chyle leak), lumbar vessels are controlled at their predictable levels, and the sympathetic fibres are found by their fixed relationship to the IMA, the lumbar veins and the aorta. Nerve sparing means recognising the anatomy before anything is divided.

AUXILIARY PROCEDURES

- **Rare at primary RPLND; needed in 23–45% of PC-RPLND. Commonest: nephrectomy, then vascular.** Larger residual mass → greater need.

Nephrectomy

- Nephrectomy is the **commonest auxiliary procedure (5–31%)**. The testes' primary lymphatic drainage lies at the great vessels near the renal hilum, and **left-sided tumours drain to the para-aortic/left hilar region**, so the **left kidney is most at risk**. Most adjunctive nephrectomies are left-sided.
- Risk is higher in **salvage, desperation and reoperative RPLND**. The incidence has **declined over three decades**.
- Need rises with histology: **fibrosis < teratoma < viable cancer**.
- **Strongest predictor: residual post-chemotherapy mass >10 cm** (odds ratio about 9). The highest-risk combination is a **left-sided residual mass >10 cm**.
- **Why it matters:** patients may receive **4–8 cycles of cisplatin**; chronic kidney disease could limit future systemic therapy, though young patients' **renal reserve usually avoids dialysis**.

TABLE 80.1 RISK FACTORS AND INDICATIONS FOR NEPHRECTOMY AT POSTCHEMOTHERAPY RETROPERITONEAL LYMPH NODE DISSECTION (Macleod 2016; Cary 2013; Djaladat 2012; Heidenreich 2009; Stephenson 2006; Nash 1998)

Category	Risk factors / indications reported
Tumour and anatomy	Retroperitoneal mass size; left-sided hilar mass ; encasement of renal vessels or ureter; involvement of renal structures; venous thrombus ; primary tumour site
Treatment setting	Salvage chemotherapy; salvage, desperation and redo RPLND ; late relapse; elevated markers
Patient	Older age; comorbidity; poor renal function
Era	Earlier year of surgery
Incidence	4.6–19% across these series. The text quotes 5–31% from the wider literature (different series and eras).

Not all studies formally tested predictors, because of small numbers.

- **What the table tests:** nephrectomy is driven by **bulk, side (left) and a hostile treatment setting**, not by histology alone.
- **Trap:** it is the **left** kidney that is at risk (left hilar drainage), and the concern is **future cisplatin eligibility**, not immediate renal failure.

Major Vascular Reconstruction

- **Plan for vascular reconstruction in every PC-RPLND:** fibrosis encases the great vessels, and **subadventitial aortic injury can cause fatal haemorrhage.**
- High-volume series need vascular procedures in **about 9–15%**. The aorta is involved most often, then the IVC, iliac vessels and renal vessels.
- **Small wall injury → bovine/porcine pericardial patch**, avoiding bypass. **Complete vessel resection → polytetrafluoroethylene (PTFE) or Dacron graft** (Fig. 80.6).
- Tumour can **invade the vessel wall with thrombus**: intraluminal thrombus in **about 6% at PC-RPLND**, almost all in the **IVC and/or renal vein**; aortic thrombus is exceptional.
- **Fig. 80.6:** A autologous patch graft of the IVC · B synthetic IVC segment with the left renal vein re-anastomosed · C synthetic graft replacing the aorta and common iliac arteries.

Inferior Vena Cava Resection

- Options: **interposition bypass graft, patch venoplasty, or lateral venorrhaphy/primary repair.**
- **Primary repair is preferred** when it will not cause significant narrowing, because the **low-pressure IVC thromboses more readily** than the arterial system.
- **Unstable patient with life-threatening bleeding → caval ligation or lateral venorrhaphy.**
- Historically **5–10% of PC-RPLND** needed caval wall or whole-IVC resection (fibrosis, invasion, thrombus). Predictors are **mass size and International Germ Cell Consensus Classification Group (IGCCCG) intermediate/poor risk**. **Mass >5 cm: about 20% need IVC intervention vs about 3% if <5 cm.**
- **Ligation** (long the Indiana standard) → **chronic venous sequelae very rare** in this young population. **Reconstruction rarely thromboses and reduces lower-limb oedema compared with acute ligation.**
- **Rule: expected narrowing <25% → primary repair/lateral venorrhaphy; >25% → patch venoplasty or interposition graft.** A patch suits cases where the full circumference need not be resected (bovine/porcine pericardium, or cadaveric vein if xenograft is unwanted).

Aortic Reconstruction

- Aortic dissection is dangerous in PC-RPLND, **especially for seminoma.**
- **Small injury → primary repair. Significant injury → almost always a graft.** There is no ligation option, unlike the IVC.
- **A multidisciplinary (vascular) team must be immediately available.**
- Commonest injury: **subadventitial dissection of a densely adherent mass → PTFE or Dacron replacement.**
- **Long duodenal serosal injury with aortic injury → graft**, to reduce the risk of delayed **aortoduodenal fistula.**
- About **1% need aortic replacement**, mostly after salvage chemotherapy or at desperation RPLND; a third were disease-free, with no graft complications. **Aortic resection is justified for complete resection** when systemic options are few.

IVC VS AORTA AT PC-RPLND

	IVC	Aorta
Pressure / thrombosis	Low pressure → grafts thrombose more readily	High pressure
Preferred repair	Primary repair / lateral venorrhaphy if <25% narrowing; patch or graft if >25%	Primary repair for small injuries only
Ligation acceptable?	Yes (unstable patient; long-term sequelae rare)	No — significant injury needs a graft

	IVC	Aorta
Graft material	PTFE; pericardial or cadaveric vein patch	PTFE or Dacron
Frequency	Caval resection historically 5–10%	Replacement about 1%

Hepatic Resections

- **IGCCCG (1997): non-pulmonary visceral metastases = poor risk** (about 48% 5-year survival). About **6%** of advanced cases have liver metastases.
- Addressed after primary, salvage or third-line chemotherapy: **all residual disease is resected, including imaged liver lesions.**
- **Liver and retroperitoneal histology are often discordant (about half)**, and most liver specimens show necrosis only. **Retroperitoneal necrosis is highly predictive of hepatic necrosis (about 94%)**, yet about 1 in 5 with retroperitoneal fibrosis only still have teratoma or viable GCT in the liver. So **residual liver masses without marker-positive disease should be resected or ablated.**
- Resection vs ablation: resection is more morbid, with **equal recurrence, except higher after percutaneous ablation.** So a **residual lesion <3 cm not safely resectable → open microwave ablation at RPLND** (limited GCT data). Small lesions in difficult sites may be biopsied and observed.

Pelvic Resections

- Rare outside **prior groin surgery, repeat RPLND or late relapse**; mostly excised with PC-RPLND.
- Histology: **teratoma in over half**, viable GCT in about a fifth, sarcomatous transformation, necrosis in about a sixth. The **fibrosis rate is less than half that of PC-RPLND specimens**, so **resect completely whenever feasible.** Surgery cures most patients.

Management of Supradiaphragmatic Disease

- **10–20%** have supradiaphragmatic disease at presentation or later. Spread is **haematogenous → lung or lymphatic → mediastinal and cervical nodes.** About **80% of mediastinal metastases are lower-middle mediastinal.** Anterior mediastinal disease suggests a primary mediastinal GCT.
- **Retroperitoneal–thoracic histology discordance: 25–50%.**
- After thoracic resection long-term survival is common, **equal for lung and mediastinal sites.** Residual pathology: teratoma in about half, necrosis in about a fifth, persistent NSGCT, and **degenerative non-germ cell cancer in about 10%.** Predictors of survival: **older age, non-germ cell cancer in the testis, residual pathology.** → **Resect residual thoracic disease >1 cm.**
- Thoracotomy series: **retroperitoneal necrosis predicted thoracic necrosis in about 89%.** → **Defer thoracic management until after RPLND.** If the retroperitoneum shows fibrosis only, **observing the chest may be warranted.** Involve a multidisciplinary team.

Resection of Retrocrural Disease

- Hard to reach from the midline. Options: **thoracoabdominal, left thoracotomy, thoracoscopic ± robotic;** thoracotomy's morbidity favours combined approaches that avoid it.
- **Transabdominal transdiaphragmatic sequence (Fig. 80.7):** standard RPLND via laparotomy → divide the gastrohepatic ligament → incise **parallel to the right crus through the muscular diaphragm** → lower posterior mediastinum. **Masses above T8 → extend with partial sternal division; full sternotomy** for subcarinal, pulmonary or anterior mediastinal disease. It gives excellent survival with low complication rates.

- **Retrocrural dissection at the same sitting is reasonable. Extensive thoracic resection** (pulmonary metastasectomy, mediastinal resection) → **staged procedure**, because of prolonged anaesthesia and higher complication risk.

EXTRA-RETROPERITONEAL RESIDUAL DISEASE: CONCORDANCE AND RULE

Site	Concordance with retroperitoneum	Predominant histology	Rule
Liver	About half discordant ; retroperitoneal necrosis → hepatic necrosis about 94%	Necrosis	Marker-negative residual lesions → resect or ablate ; <3 cm unresectable → open microwave ablation
Thorax	25–50% discordant ; retroperitoneal necrosis → thoracic necrosis about 89%	Teratoma, necrosis	RPLND first ; resect >1 cm unless the retroperitoneum shows fibrosis only
Pelvis	—	Teratoma ; fibrosis uncommon	Complete resection whenever feasible
Retrocrural	—	—	Same sitting (transdiaphragmatic); stage it if extensive thoracic surgery is needed

KEY POINTS: AUXILIARY PROCEDURES

- **Nephrectomy** commonest (left-sided, large masses).
- Plan for **aortic replacement**; **primary IVC repair** preferred to graft.
- Discordance common: **chest after RPLND**; resect >1 cm unless retroperitoneum fibrosis only.
- Retrocrural at RPLND; **extensive thoracic** → **staged**.

Concept — *Chemotherapy response differs between organs, so histology at one site never guarantees histology at another. The retroperitoneum is cleared first because it is the best, though imperfect, predictor of what remains elsewhere, and the surgeon must be ready to take kidney, cava or aorta to make the resection complete.*

SURGICAL DECISION MAKING

Management of Clinical Complete Remission to Induction Chemotherapy

- About 70% of patients with clinical stage II or higher GCT treated with cisplatin achieve a **complete radiographic (no residual mass >1 cm) and serological remission**. The options are **observation or PC-RPLND**. This remains controversial.

COMPLETE REMISSION: OBSERVATION VS PC-RPLND

	Observation	PC-RPLND for all
Argument	Excellent long-term recurrence-free survival (RFS) and cancer-specific survival (CSS); retroperitoneum-only relapses are all salvaged by RPLND	Even a residual mass ≤1 cm harbours teratoma in about a fifth and viable GCT in about 1 in 20
View of microscopic teratoma	Indolent in most cases	Risk of growing teratoma syndrome, late relapse, malignant transformation , all chemoresistant and hard to salvage

	Observation	PC-RPLND for all
Who does it	Indiana and many major centres (radiological ≤ 1 cm and serological complete response)	Some high-volume centres

TABLE 80.2 MANAGEMENT OF PATIENTS EXPERIENCING A CLINICAL COMPLETE REMISSION TO INDUCTION CHEMOTHERAPY (Ehrlich 2010; Kollmannsberger 2010; Karellas 2007)

Series	Management	Good risk	Follow-up (years)	Disease-free survival (DFS)	CSS
Ehrlich 2010	Observation	77%	15.5 (text: 15-year)	91% (text: RFS 90%)	97%
Kollmannsberger 2010	Observation	94%	4.3 (text: median 4.5)	94% (text: RFS 93.8%)	100%
Karellas 2007	PC-RPLND	98%	3	97%	Not reported

The text and the table differ slightly on endpoint (RFS vs DFS), follow-up summary (median vs reported) and rounding; both values are shown.

- **What it tests:** non-randomised series using opposite strategies give **similar, excellent results**. Both work if patients are well selected.
- **Pattern and trap:** the cohorts are overwhelmingly **good risk** with unequal follow-up. The table **does not show equivalence**, and it cannot justify either policy in intermediate/poor-risk disease.
- **Predictors of necrosis/fibrosis only at PC-RPLND** (meta-analysis): **no teratoma in the orchidectomy specimen; seminomatous elements; normal pre-chemotherapy AFP/ β -hCG; elevated LDH; >50% shrinkage during chemotherapy; small residual mass (<2 cm).**
- **Vergouwe nomogram (2007)** uses similar variables to give a percentage risk of necrosis/fibrosis only.
- The debate will persist until it is shown that (1) PC-RPLND **prevents cancer deaths** after complete response, (2) the **number needed to treat** is low enough to justify the morbidity, and/or (3) a **robust predictive tool** (biomarker, nomogram) is available for routine use.

Use of Modified Templates in Primary Retroperitoneal Lymph Node Dissection

- **Historical RPLND:** bilateral infra hilar **plus interiliac region to the common iliac bifurcation**, often **bilateral supra hilar**, via thoracoabdominal incision: high morbidity, **most anejaculatory**, accepted before curative chemotherapy existed.
- **Revolution, 1970s–1980s:** curative cisplatin chemotherapy (Einhorn and Donohue, 1977) + recognition of **side-specific lymphatic spread + nerve-sparing techniques** → lower-morbidity modified templates.
- **Ray (1974):** the first templated modifications, focused on the primary landing zones of right- vs left-sided tumours.
- **Donohue mapping study (1982):** left tumours → **left para-aortic**; right tumours → **interaortocaval and precaval**. Contralateral and supra hilar spread is rare but **increases with retroperitoneal bulk**; **interiliac spread is very rare**.
- **Weissbach and Boedefeld (1987):** a reduced left template (para-aortic and pre-aortic) in non-bulky pathological stage II, with **intraoperative frozen section** to trigger a full template.
- **Consensus: omit interiliac and supra hilar regions in the absence of bulky disease.** Omitting supra hilar dissection ↓ **chylous ascites, renal injury, and pancreatic/duodenal injury**. Omitting interiliac **plus** contralateral dissection helps **preserve antegrade ejaculation**. **Omitting the contralateral side remains controversial**.
- **Against unilateral templates:** a high-volume series of primary RPLND in clinical stage I/IIA applied a full template (supra hilar usually omitted). Depending on the template, **3–23% of node-positive patients had a positive node outside the modified template, more with right-sided tumours**. The authors recommended **full bilateral infra hilar nerve-sparing RPLND** as standard for clinical stage I/IIA.

- **No randomised trial.** Both achieve **CSS approaching 100%** (Table 80.3); comparison is confounded by **differing adjuvant policies**, so there is no one-size-fits-all template.
- **Fig. 80.8:** modified right and left templates (A) vs bilateral template (B), bounded by the **renal vessels above, ureters laterally and common iliac arteries below**.

TABLE 80.3 SELECTED PRIMARY RPLND SERIES (Donohue 1993a; Stephenson 2005; Hermans 2000; Nicolai 2004)

Endpoint	Range across the four series
Histologically node-positive (pN+)	about 20–30%
Recurrence if node-negative (pN0)	about 7–12%
Recurrence if pN+ managed by RPLND alone	about 23–34%
CSS	98.8–100% (follow-up 3.8–7.2 years)
Adjuvant policy	Stephenson: many pN2 given adjuvant · Donohue 1993a: pN+ randomised to observation vs adjuvant · Hermans: pN1 and most pN2 observed; chemotherapy at relapse or for pN3

- **What it tests:** primary RPLND cures almost everyone, whatever the template. About **one-quarter to one-third of pN+ patients relapse after surgery alone**, and chemotherapy salvages them.
- **Trap:** identical CSS **cannot rank templates**, because adjuvant chemotherapy use differed between series.

Use of Modified Templates in Retroperitoneal Lymph Node Dissection After Chemotherapy

- **Standard of care:** resect the residual mass + **full bilateral infrahilar RPLND**. This has been the standard since the **first PC-RPLND report (Donohue, 1982): contralateral crossover is frequent with bulky initial disease** and viable tissue cannot be recognised intraoperatively. The cost is morbidity and ejaculatory dysfunction where nerve sparing is not feasible.
- Modified templates after chemotherapy (Table 80.4): extra-template nodes vary widely, yet **recurrence stays low**.
- **Candidate criteria for modified-template PC-RPLND (all three):**
 1. **Well-defined lesion ≤ 5 cm, confined to the primary landing zone** on pre- and post-chemotherapy imaging (Fig. 80.9);
 2. **Normal post-chemotherapy STMs;**
 3. **IGCCCG good or intermediate risk.**
- With these criteria: **in-field recurrence 0–4%, CSS 98–100%** (up to about 10 years), **antegrade ejaculation >85%**. No randomised comparison exists, so **full bilateral dissection remains standard**.
- **Fig. 80.9 (CT):** A precaval mass → candidate for a **modified right** template · B para-aortic mass → candidate for a **modified left** template · C → needs an **extensive bilateral** PC-RPLND.

TABLE 80.4 STUDIES EVALUATING THE USE OF MODIFIED UNILATERAL TEMPLATES IN POST-CHEMOTHERAPY RETROPERITONEAL LYMPH NODE DISSECTION (Wood 1992; Herr 1997; Rabbani 1998; Ehrlich 2006; Beck 2007b; Steiner 2008; Carver 2007a; Heidenreich 2009; Cho 2017)

Endpoint	Range across series (where reported)
Positive nodes outside the modified template	about 7–32%
Outside-template positive nodes with macroscopic disease	about 2–19%
In-field retroperitoneal recurrence after bilateral RPLND	0–4%
In-field retroperitoneal recurrence after unilateral RPLND	0–3% (one outlier of 9% occurred after tumourectomy only)

Endpoint	Range across series (where reported)
Ejaculation preserved with templates	50–94%
CSS	96–100%; one outlier of 89%
Follow-up	2.6–10.4 years

- **What it tests:** node-positive tissue outside the template is common, yet recurrence is rare in selected patients. Selection, not the template itself, protects oncological control.
- **Trap:** good results in these series do **not** make modified templates the PC-RPLND standard. They apply only to patients meeting all three criteria.

TEMPLATE STANDARD: PRIMARY VS POST-CHEMOTHERAPY RPLND

	Primary RPLND	PC-RPLND
Default template	Modified unilateral or full bilateral infrahilar nerve-sparing; both acceptable, no randomised trial	Residual mass + full bilateral infrahilar, nerve-sparing where feasible
Modified template allowed when	Low-stage, low-volume disease	Only if ≤ 5 cm in landing zone + normal STMs + IGCCCG good/intermediate
Main oncological risk	Missed extra-template nodes (right > left)	Contralateral crossover with bulky disease; viable tissue unrecognisable

Adjuvant Chemotherapy for Pathological Stage II Disease at Primary Retroperitoneal Lymph Node Dissection

- pN0 → observation, with no further therapy.

PN1–N2 AFTER PRIMARY RPLND: OBSERVATION VS ADJUVANT CHEMOTHERAPY

	Observation (treat at relapse)	Adjuvant chemotherapy (usually 2 cycles)
For	About 70% are cured by surgery alone; relapses are almost all salvaged by chemotherapy	Virtually eliminates relapse in pN1/pN2
Against	Salvage is more intensive : at least one extra cycle of bleomycin, etoposide and cisplatin (BEP), or two extra cycles of etoposide and cisplatin (EP)	About 70% treated unnecessarily and exposed to toxicity

- The ideal is to **confine adjuvant therapy to a high-risk group**, but no reliable predictor exists.
- **Nodal bulk:** in the observation arm of the **Williams randomised trial (1987)**, relapse was about 40% (microscopic), 53% (macroscopic <2 cm) and 60% (>2 cm), but the differences were **not significant**. Retrospective data conflict: some show higher relapse with larger burden, others no difference between pathological IIA and IIB, probably because of selection and confounding.
- **Other tested predictors** (positive-node ratio, metastatic histology, **extranodal extension**) are **not reliable**.
- **One exception: teratoma only in the retroperitoneum** → very low relapse → observation alone.

ADJUVANT DECISION BY NODAL STAGE

Stage	Approach
pN0	Observation
Teratoma only	Observation
pN1	Observation in reliable patients (broad consensus)
pN2	Controversial: two cycles of adjuvant chemotherapy vs observation; individualise

KEY POINTS: SURGICAL DECISION MAKING

- **Complete response** → **observe** (generally).
- Modified templates: primary low-volume; PC only under strict criteria.
- **Teratoma only** → **no adjuvant**; **pN1 observe**; **pN2 individualise**.

Concept — *How far to operate depends on how predictable the disease is. In untreated low-volume disease the landing zones are reliable enough to narrow the field and to observe after resection. After chemotherapy, bulk, crossover and unrecognisable residual tissue make the map unreliable, so the default reverts to the full bilateral template.*

HISTOLOGIC FINDINGS AT POST-CHEMOTHERAPY RETROPERITONEAL LYMPH NODE DISSECTION AND SURVIVAL OUTCOMES

- Three findings at PC-RPLND: **fibrosis/necrosis about 40%, teratoma about 45%, viable GCT about 15%**. Viable GCT has become less frequent as chemotherapy regimens have been refined.

Outcomes by Histology

- CSS is **excellent after teratoma or fibrosis**; the concern is **viable GCT** (Table 80.5).
- Histology after **BEP ×3 vs EP ×4** (good risk) is debated, but **long-term survival is similar**. Series vary by era, pretreatment risk and follow-up.

Fibrosis/Necrosis

- **CSS and RFS approach 95%**. Chemotherapy has rendered all malignant cells non-viable, and **subclinical deposits have most likely been cleared as well**.

Teratoma

- **RFS 80–90%, CSS 85–95%**. Recurrence risk is higher with **somatic-type malignancy** and **mediastinal primaries**.

Viable Malignancy

- Persistent viable disease means **primary resistance to chemotherapy**. Long-term survival is **50–70%**.

FIZAZI PROGNOSTIC MODEL FOR VIABLE MALIGNANCY AT PC-RPLND (MULTI-INSTITUTIONAL, 2001/2008)

Risk group	Adverse factors present	5-year progression-free survival (PFS)	5-year overall survival (OS)
Favourable	None	90%	100%
Intermediate	One	76%	83%
Poor	Two or three	38%	51%

- **The three factors:** (1) **incomplete resection**; (2) **≥10% viable malignant cells**; (3) **IGCCCG intermediate/poor risk at diagnosis**.
- **Trap:** the model is **prognostic only**. It does **not predict benefit from adjuvant chemotherapy**.

TABLE 80.5 SURVIVAL OUTCOMES BY HISTOLOGIC FINDINGS AT POST-CHEMOTHERAPY RETROPERITONEAL LYMPH NODE DISSECTION (Donohue and Foster 1994; Eggener 2007a; Carver 2006, 2007c; Maroni 2008; Loehrer 1986; Jansen 1991; Beck 2009; Fox 1993; Donohue 1998; Fizazi 2001; Spiess 2007; Kundu 2010)

Histology	RFS (table range)	CSS (table range)	Text summary
Fibrosis	92–95%	85–93%	CSS and RFS approach 95%

Histology	RFS (table range)	CSS (table range)	Text summary
Teratoma	61–89%	77–98%	RFS 80–90%, CSS 85–95%
Viable malignancy	31–64%	43–73%	Long-term survival 50–70%

The lowest table values come from cohorts given salvage chemotherapy before PC-RPLND, and from older-era series. The text quotes contemporary figures, which is why the two differ.

- **What it tests:** survival falls steeply at one point, **viable GCT**. Fibrosis and teratoma behave almost alike.
- **Trap:** in any histology group, **prior salvage chemotherapy lowers survival**. Always check whether a series is first-line or salvage.

Adjuvant Chemotherapy

- **Debated.** Historically **two cycles** of variable regimen; associated with **better PFS but not OS**.
- Individualise by **original IGCCCG risk group, pathology (percentage viable) and completeness of resection**.
- **Fibrosis/necrosis or teratoma → no adjuvant chemotherapy.**

GUIDELINE BODIES: ADJUVANT CHEMOTHERAPY AFTER VIABLE GCT AT PC-RPLND

	National Comprehensive Cancer Network (NCCN)	European Association of Urology (EAU)
Recommend adjuvant	Incomplete resection; IGCCCG intermediate or poor risk	Incomplete resection
Do not recommend	—	IGCCCG good risk with <10% viable tumour (complete resection)
After salvage/second-line chemotherapy	Two further adjuvant cycles not recommended: no improvement in outcome	
Evidence	No randomised data, even after single-line chemotherapy	

KEY POINTS: HISTOLOGIC FINDINGS AT PC-RPLND AND OUTCOMES

- **Fibrosis/teratoma about 90% vs viable GCT 50–70% long-term survival.**
- **Adjuvant for viable GCT individualised; none for fibrosis or teratoma.**

Concept — At PC-RPLND, histology is the prognosis. Fibrosis and teratoma mean chemotherapy and surgery have each done their job. Viable GCT means chemoresistance, and its outlook depends on how completely it was removed, how much remained, and how bad the disease was at the start.

POST-CHEMOTHERAPY RETROPERITONEAL LYMPH NODE DISSECTION IN HIGH-RISK POPULATIONS

Salvage Retroperitoneal Node Dissection

- Performed in **particularly high-risk, partly platinum-resistant** disease, typically after **at least 7–8 cycles** of first- and second-line chemotherapy to normalise markers.
- Viable malignancy is about **15–20% after first-line chemotherapy** but **substantially higher after salvage (at least double)**, with worse OS. In one salvage series: fibrosis/necrosis 50%, teratoma 23%, viable 27%. **OS after salvage RPLND 60–75%.**

- **After third-line high-dose chemotherapy (HDCT) with stem cell transplantation:** extremely rare. Necrosis about a quarter, teratoma about a third, **viable cancer about 40%; 5-year OS about 70%**. This needs careful selection and multidisciplinary decision making, but **RPLND still has a significant role**.

Desperation Retroperitoneal Node Dissection

- A main indication for RPLND after any line of chemotherapy is **normalisation of STMs**. Desperation RPLND is the special case of **persistently elevated STMs despite chemotherapy**. It must **never be undertaken outside a multidisciplinary setting**, because relapse and progression risk is very high.
- Largest series: **5-year OS about 54%**. Independent predictors of poor OS: **desperation RPLND after salvage chemotherapy, elevated pre-operative β -hCG, and repeat RPLND**.
- Markers that fell after first-line chemotherapy and then **plateaued at a detectable level** → **>75% chance of fibrosis or teratoma**. Paradoxically, **fibrosis here may mean a worse prognosis**: the markers must then be coming from disease outside the retroperitoneum.
- **Normalisation of markers after RPLND** is the only variable linked to improved OS.

DESPERATION RPLND: WHEN TO OPERATE AND WHEN NOT TO

Operate	Do not operate (→ salvage or third-line HDCT)
Limited metastatic sites (ideally retroperitoneum only; may include thoracic or liver sites) that are completely resectable at one, non-staged operation , with declining or plateauing STMs	Rising markers with rapid doubling times
Chemotherapy options exhausted, with resectable disease and elevated markers	Progressive metastatic disease at multiple sites

Reoperative Retroperitoneal Node Dissection

- Usually a result of **technical failure of the first operation**, which left residual disease at primary or post-chemotherapy RPLND. This is why RPLND belongs with **experienced high-volume surgeons**. **Laparoscopic or robotic RPLND must obey the same rules of complete dissection** as open surgery.
- **Volume:** most RPLNDs in board recertification logs are done by urologists who have **performed only one**. **>2 per year = top 25%; 3 per year = top 10%**.
- In the largest series, **local recurrence was predicted by incomplete lumbar vessel division** at the first operation and by **teratoma** in the reoperative specimen. **Worse outcome: active cancer at reoperation, M1b stage, salvage chemotherapy before reoperation**.
- Most recurrences lie in the **primary landing zone**, and **about half are retroaortic or retrocaval**, i.e. left behind. Viable GCT with **somatic-type malignancy in 15–20%**. Complication rates are higher and cancer-specific outcomes worse.
- → **Thorough dissection with complete mobilisation of the great vessels at the first RPLND is mandatory**.

Late Relapse

- **Definition:** recurrence **more than 24 months** after primary treatment with an initial complete response, outside observation. **Rare: 2–4%**.
- **Commonest site: retroperitoneum**. **Yolk sac** is often the dominant histology, and **somatic-type malignancy** is more frequent. Both are **chemoresistant**.
- → **Resectable disease: RPLND first**. **Chemotherapy is reserved for widespread or unresectable disease**, to downsize it towards resectability.
- OS after surgery is **nearly 60%**, so select cases carefully. **Adverse factors:** viable GCT, somatic-type malignancy, prior salvage chemotherapy, incomplete resection.

TABLE 80.6 POST-CHEMOTHERAPY RETROPERITONEAL LYMPH NODE DISSECTION IN HIGH-RISK POPULATIONS (Fox 1993; Donohue 1998; Eggener 2007a; Rick 2004; Cary 2011; Ravi 1998; Albers 2000; Beck 2005c; Ong 2008; McKiernan 2003; Sexton 2003; Heidenreich 2003; Willis 2007; Pedrosa 2014; Baniel 1995; George 2003; Dieckmann 2005; Sharp 2008)

Setting	Teratoma	Fibrosis	Viable malignancy	CSS or OS
<i>First-line (reference)</i>	<i>about 45%</i>	<i>about 40%</i>	<i>about 15%</i>	<i>about 90% (fibrosis/teratoma)</i>
Salvage	about 21%	about 51%	28–55%	37–74%*
After HDCT	16–34%	27–38%	39–46%	65–71%
Desperation	11–34%	12–28%	47–64%	54–69%
Redo	33–67%	9–44%	22–56%	56–94%
Late relapse	19–71%	0–3%	78–81%	57–75%

*The lowest salvage value counts only patients with viable malignancy. Text vs table: after HDCT the text reports necrosis 26%, teratoma 34%, viable 38% and 5-year OS 70% (a later, larger data cut of the same institution's series), and the table 27%/34%/39%/71%. For desperation RPLND the text gives 116 patients with 5-year OS 53.9% and the table 114 patients (different denominator). For salvage, the text's single series (viable 27%) falls at the low end of the table.

- **What it tests:** the more chemotherapy has failed, the **less fibrosis and the more viable tumour**. Late relapse is the extreme case, **almost never fibrosis**.
- **Trap:** redo RPLND can show a **high CSS in some series** despite a high viable fraction. The main cause of redo surgery is technical, not biological.

KEY POINTS: PC-RPLND IN HIGH-RISK POPULATIONS

- Salvage: viable GCT **at least double**; multidisciplinary decision.
- Desperation RPLND **never outside a multidisciplinary setting**.
- Redo RPLND is avoidable: complete first dissection with great-vessel mobilisation.

Concept — Each high-risk setting is defined by chemotherapy failure. The fibrosis fraction therefore falls and chemoresistant histology rises until surgery is the only effective therapy left, which is why completeness of resection and multidisciplinary selection decide the outcome.

SURGICAL OUTCOMES, FUNCTIONAL CONSIDERATIONS, AND COMPLICATIONS OF RETROPERITONEAL LYMPH NODE DISSECTION

Lymph Node Counts

- Proposed as a **surgical quality measure**, as in other cancers; results **mixed** (higher counts linked to node positivity in one series, not in others).
- Counts also depend on **how pathology submits and processes tissue**, not only on the surgeon.
- National datasets link **higher counts to better oncological outcomes** and to **academic/high-volume centres**. Whether the count is a valid surrogate for quality is **unproven**.

Retroperitoneal Lymph Node Dissection and Fertility

Fertility in Patients Undergoing Retroperitoneal Lymph Node Dissection

- RPLND **does not affect testicular function or spermatogenesis**, but **subfertility is common at diagnosis: abnormal semen in 40–60%** of newly diagnosed men. This confounds paternity rates after treatment.
- Live birth after orchidectomy and primary RPLND is about **one-third** (small cohort).

Ejaculatory Dysfunction and Retroperitoneal Lymph Node Dissection

- **Physiology: seminal emission** (smooth-muscle contraction of the vasa, seminal vesicles and prostate, and prostatic secretion) and **bladder-neck closure** (which prevents retrograde ejaculation) both depend on the **L1–L4 postganglionic sympathetic fibres**. These coalesce **at or below the IMA in the hypogastric plexus** and run caudally in the **interiliac space over the sacral promontory**.
- Pre-modification RPLND removed these fibres → **anejaculation**.

THE TWO MODIFICATIONS THAT RESTORED EJACULATION

	Modified unilateral template (no nerve dissection)	Prospective nerve sparing
Principle	Selected patients: dissection confined to one side leaves the contralateral fibres untouched	Fibres identified and preserved during dissection (Jewett and Torbey, 1988)
Antegrade ejaculation	Early 75–87% ; refined 97%	90–100%
Caveat	Requires suitable low-volume disease	Early series had transient postoperative anejaculation , resolved; now absent, probably by avoiding traction neuropraxia

- Combined, these give **antegrade ejaculation in >90%**, especially in the primary setting. **Paternity after nerve-sparing primary RPLND is up to 75%**. Paternity after PC-RPLND is **not established**, because chemotherapy has a prolonged effect on spermatogenesis.
- **No oncological cost**: only one retroperitoneal recurrence was reported across the nerve-sparing and template series.

Complications of Retroperitoneal Lymph Node Dissection

- Overall complications: **primary RPLND 10.6–24%**; **PC-RPLND up to 56%**, less at experienced centres.
- **Risk factors**: tumour size, IGCCCG intermediate/poor risk, elevated markers, prior RPLND, longer operative time, auxiliary procedures.
- The **mean number of RPLNDs per board-certified urologist per year is 1**, which explains much of the variability in reported outcomes.

TABLE 80.7 COMPLICATIONS OF RETROPERITONEAL LYMPH NODE DISSECTION (Primary: Baniel 1994; Heidenreich 2003; Subramanian 2010. PC-RPLND: Baniel 1995; Subramanian 2010)

Complication (%)	Primary RPLND (3 series)	PC-RPLND (2 series)	Text range
Overall	10.6–24	20.7–32	Primary 10.6–24; PC up to 56
Major	3–8.2	8 (not studied in one)	
Mortality	0	0.8–1	PC <1
Major pulmonary	0.8–1.9	3.1–5.1	PC 3–5
Minor pulmonary	0.2–3.6	3.1–5.1	
Chylous ascites	0.2–2.1	2	Primary 0.2–2.1; PC 2–12
Symptomatic lymphocele	0–1.7	1–1.7	≤1.7
Ileus	2.1–17.9	2.2–20.8	Up to 26
Wound infection	0.9–5.4	4–4.8	
Pulmonary embolism	0–0.9	0.1–3.1	Venous thromboembolism: primary <1; PC 0.1–3.1

Complication (%)	Primary RPLND (3 series)	PC-RPLND (2 series)	Text range
Ureteric injury	0.2–0.9	0–0.9	
Small bowel obstruction	0.4–2.7	1.8–2.3	
Postoperative haemorrhage	0–0.8	0.3–1	

The text ranges are wider than the table's because they draw on additional series.

- **What it tests:** the complications that **rise most after chemotherapy** are **pulmonary** (bleomycin), **mortality** and, in the wider literature, **chylous ascites**. Ileus is common in both settings.
- **Trap:** the commonest complications are **ileus and wound infection**, not chylous ascites, which is rare after primary RPLND.

Pulmonary Complications

- Major pulmonary complications are rare after primary RPLND but occur in **3–5% after PC-RPLND**, falling as teams become more aware of bleomycin.
- **Bleomycin-treated patients are at increased risk of acute respiratory distress syndrome (ARDS) and prolonged ventilation.**
- **Prevention:** avoid aggressive intravenous fluid intra- and postoperatively; keep **FiO₂ as low as safely possible**; use a bleomycin-aware anaesthetic team. Warn recovery and ward staff that **tachycardia is common after RPLND and is not on its own a trigger for fluid**. Use nasal oxygen judiciously and **accept oxygen saturation of 88–90%**.

Ileus

- Up to **26%** after primary or PC-RPLND. The **extraperitoneal approach may significantly reduce it**. No routine nasogastric tube; some **enteral feeding on day 1** unless circumstances prevent it.

Lymphocele

- Mostly subclinical; **symptomatic ≤1.7%: fullness or flank pain** (ureteric compression); CT shows a **thin-walled cyst in the bed**.
- **Air within it and/or rim enhancement → suspect infection.**
- Symptomatic or infected → **percutaneous drainage**. Antibiotics only if infected, and then consider leaving a drain. **Prevention: clip or ligate lymphatic channels.**

Chylous Ascites

- Lipid-rich lymph in the peritoneal cavity: **0.2–2.1% after primary RPLND vs 2–12% after PC-RPLND**.
- **Increasing girth, fullness, anorexia, weight gain; dyspnoea** if large. Diagnosis: suspicion plus CT.
- **Paracentesis: milky fluid, protein >3 g/dL, triglyceride 2–8 × serum.**
- **Suprahilar dissection raises the risk** by disrupting the **cisterna chyli**, which is variable: at **L1–L2, at or just above the renal vessels**, most consistently **behind the aorta under the left crus**. A distinct cisterna is present in only **about half**.
- **Management ladder:** paracentesis without a drain → if rapid reaccumulation, a drain capped for interval release → **very low-fat, high-protein diet + medium-chain triglycerides + intramuscular octreotide** (proven for chyle leak after hepatopancreaticobiliary surgery, not in urology) → failure: **nil by mouth + parenteral nutrition** → persistent high volume: observe, or **invasive last resort: lymphangiography/lymphoscintigraphy ± percutaneous embolisation, peritoneovenous shunt** (resolved about 90% of refractory cases in a small series), or surgical exploration.
- Invasive options are for leaks **not improving over days to weeks**, or with **unmanageable electrolyte and protein loss**. After resolution, continue a **low-fat diet for at least 4 weeks**.

Venous Thromboembolism

- Venous thromboembolism (VTE) is low in this young, healthy population: **<1% after primary RPLND, 0.1–3.1% after PC-RPLND.**
- **Sequential compression devices from the holding area;** mobilise on **day 0**, with long periods out of bed.
- Pharmacological prophylaxis is **unstudied in RPLND** but proven in general surgery. Deep vein thrombosis (DVT) occurs in **about 13% during first-line bleomycin chemotherapy.** → **Pharmacological prophylaxis advocated**, especially with prior VTE, hypercoagulable states, obesity or prior chemotherapy (particularly bleomycin).

Neurologic Complications

- Rare peripheral nerve injury from **positioning or retractors (femoral neuropraxia)**; prevent by careful positioning.
- **Bulky mediastinal or retroperitoneal disease after chemotherapy** → **increased risk of paraplegia.**

Mortality

- **Primary RPLND: exceptionally rare. PC-RPLND: <1%,** most often from **pulmonary embolism, duodenal fistula, myocardial infarction or chylous ascites.**

KEY POINTS: SURGICAL OUTCOMES, FUNCTIONAL CONSIDERATIONS AND COMPLICATIONS

- **Antegrade ejaculation expected** after most primary RPLNDs.
- **PC-RPLND: bleomycin lung** — FiO₂, fluids, anaesthesia.
- **Chylous ascites** prevented by lymphatic ligation.

Concept — Morbidity follows the chemotherapy history. Bleomycin lung, fibrotic planes and bulk turn a low-morbidity staging operation into a major one, so prevention (low FiO₂, restrictive fluids, lymphatic ligation, thromboprophylaxis) is planned before the incision. Ejaculatory function depends only on whether the L1–L4 fibres are recognised and spared.

RETROPERITONEAL LYMPH NODE DISSECTION IN UNIQUE SITUATIONS**Post-Chemotherapy Retroperitoneal Node Dissection for Seminoma**

- **Pure seminoma is exquisitely chemosensitive: complete response 70–90%** with platinum, even when widely metastatic, so residual masses are uncommon.
- Surgery is hard and rarely rewarding: **dense desmoplastic fibrosis, viable tumour only about 10%, no teratoma, more complications than NSGCT.**
- **The 3 cm rule: residual masses <3 cm were all fibrosis, while >3 cm harboured viable tumour or teratoma in about 30%. → Observe all masses <3 cm;** resection of >3 cm was recommended, but **70% of those are fibrosis**, so many still undergo morbid surgery. Indiana data found **no correlation between residual size and relapse** and advised observing all masses unless they progress serologically or radiologically.
- → **FDG-PET decides for masses >3 cm.**

FDG-PET FOR RESIDUAL SEMINOMA MASSES — SEMPET STUDY (DE SANTIS, 2004)

Residual mass	Specificity	Sensitivity	Positive predictive value (PPV)	Negative predictive value (NPV)
>3 cm	100%	80%	100%	96%

Residual mass	Specificity	Sensitivity	Positive predictive value (PPV)	Negative predictive value (NPV)
≤3 cm	74%	70%	37%	92%

- All masses >3 cm and 95% of those ≤3 cm were correctly classified.
- **<3 cm:** fibrosis is historically almost 100% and a positive scan is unreliable (**low PPV**), so **PET has little or no role**. **>3 cm:** PET is a **useful adjunct** for choosing between observation and treatment.

GUIDELINE BODIES: IMAGING OF POST-CHEMOTHERAPY SEMINOMA MASSES

NCCN	EAU
PET for masses >3 cm; observe masses <3 cm	PET as a general tool for suspicious retroperitoneal masses after chemotherapy

- **PET-avid mass >3 cm → PC-RPLND, salvage chemotherapy, or HDCT with stem cell transplantation.**
- The choice is controversial, because **residual seminoma at PC-RPLND is ominous**: 5-year CSS about 54%, death on average about 7 months after surgery, and few patients remain continuously disease-free. **HDCT gives OS of about 92% second-line and 64% third/fourth-line** in pure seminoma, so consider it as an alternative to consolidative surgery.
- → **PC-RPLND for seminoma only for PET-avid, easily resectable masses** in very select cases where it spares HDCT morbidity, or for **chemotherapy-ineligible** patients.

RESIDUAL MASS AFTER CHEMOTHERAPY: SEMINOMA VS NSGCT

	Pure seminoma	NSGCT
Surgical threshold	>3 cm and PET-avid; observe <3 cm	Residual mass >1 cm with normal markers
Teratoma	Absent	About 45% of PC-RPLND specimens
Viable tumour	About 10%	About 15%
Surgical field	Dense desmoplastic fibrosis ; more operative complications; aortic danger	Fewer operative complications than seminoma

Post-Chemotherapy Retroperitoneal Node Dissection for Sex Cord Stromal Tumours

- Sex cord stromal tumours (SCSTs) are **<5% of testicular neoplasms: Leydig cell, Sertoli cell and granulosa cell tumours**, pure or mixed. **Adult granulosa cell tumour is the rarest testicular tumour** (fewer than 50 reported cases). Series are small.
- **Malignancy is hard to diagnose** without metastases. Features associated with malignant behaviour: **older age, size >4 cm, necrosis, mitoses >3–5 per 10 high-power fields, moderate–severe nuclear atypia, infiltrative margins/invasion of adjacent structures, lymphovascular invasion**. Some classify **two or more features as malignant**, but prospective histological classification remains difficult.
- Sertoli plus Leydig: about **0.4–0.6% of testicular tumours** (registries).

SERTOLI CELL VS LEYDIG CELL TUMOURS (NATIONAL REGISTRY DATA)

	Sertoli cell tumour	Leydig cell tumour	GCT (reference)
Stage at presentation	35% stage II/III	98.5% stage I	—
Cancer-specific mortality	32%	2%	7%
Black patients	23%	24%	3%

	Sertoli cell tumour	Leydig cell tumour	GCT (reference)
Age	Older than GCT patients		—

- **What it tests:** Sertoli cell tumours can be aggressive; Leydig cell tumours are almost always localised.
- **Trap:** SCSTs occur disproportionately in **Black patients**, in contrast with GCT.
- **Role of RPLND:** metastatic SCST does badly despite additional therapy (most stage II–III patients died; **CSS and OS about 30% in stage II**). **Metastasis is the only predictor of survival.** → **No conclusive evidence that RPLND improves survival**; it has only a limited role.

KEY POINTS: RPLND IN UNIQUE SITUATIONS

- Seminoma: **<3 cm observe**; **>3 cm PET**; PET-avid → surgery, salvage or HDCT.
- Malignant SCST: **RPLND of no proven survival benefit.**

Concept — Both unique situations break the NSGCT rules. Seminoma is chemosensitive and has no teratoma, so residual masses are mostly scar, and imaging (size, then PET) rather than routine resection decides surgery. SCSTs are refractory once metastatic, so the operation that cures NSGCT offers them little.

CONCLUSION

- Testicular cancer is one of oncology's great successes: **cure in more than 90%**. This is largely due to **cisplatin-based multiagent chemotherapy**, but **surgery remains central**.
- The urologist usually **diagnoses and guides the patient**, within a **multidisciplinary team**.
- Success is now measured as **cure with the least morbidity and treatment burden**. Surgical refinements aim to protect quality of life and avoid long-term sequelae **without losing oncological efficacy**.